



**Faculty of
Computing**

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INITIAL SYSTEM PROPOSAL Hasta Travel & Tours Car Rental System

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1.0 Introduction

This project proposal outlines the development of a Car Rental Management System for Hasta Travel and Tours. The aim of this system is to digitalize and streamline the company's car rental operations, including vehicle booking, customer management, payment tracking, and report generation. Currently, many rental processes of this company are handled manually, which can lead to inefficiencies, data redundancy and human errors. Therefore, this system aims to provide a centralized, reliable and user-friendly platform to improve business productivity and service quality.

In this proposal, we will introduce the client's existing system used to operate the car rental business, how our proposed system can assist Hasta Travel and Tours in managing their operations more efficiently and the overall objectives of the system. Currently, the client relies on manual methods and messaging applications such as WhatsApp to handle booking requests, payment confirmations, and customer inquiries. This approach often leads to delays, data mismanagement and errors due to the absence of a centralized system.

This project will be implemented across three Work-Based Learning (WBL) subjects which are Database(WBL), Software Engineering(WBL), and System Development Technology(WBL). Through this integration, We will apply theoretical knowledge into real-world practice by designing a structured database, following proper software development life cycle (SDLC) methodologies and implementing a functional web-based system.

As the project stakeholder, Hasta Travel and Tours will benefit from improved operational efficiency, better data organization and enhanced customer experience. The system will allow administrators to manage vehicles, monitor bookings, track rental history, and generate business reports effectively. Customers will also benefit from a more convenient booking process through an online platform.

Overall, this project supports the digital transformation of Hasta Travel and Tours while providing us with hands-on industry experience, making it a valuable learning and business solution.

2.0 Background Study

Hasta Travel & Tours Sdn. Bhd., also known as Car Rental UTM, is a licensed travel and car rental company based at Student Mall, Universiti Teknologi Malaysia (UTM), Skudai, Johor. It offers a wide range of vehicles, from affordable models like Perodua Axia, Bezza, and Myvi to premium options such as Toyota Alphard and Hyundai Starex, catering to both short-term and long-term rental needs. Currently, the rental process is handled manually, where customers make inquiries via WhatsApp, phone calls, or in-person visits. Staff confirm bookings by checking availability through spreadsheets or written records, while customers submit personal details and payment proof manually. On the day of rental, vehicles are collected at the office or agreed location, and after return, deposits are refunded—often taking several days. This traditional process can lead to booking errors, record-keeping issues, and delays in customer service. As the demand for car rental services grows, there is a strong need to digitalize the system to improve operational efficiency, customer satisfaction, and business scalability.

2.1.1 Car Inquiry and Reservation

The process begins when customers submit rental inquiries through whatsapp and walk-in. Upon receiving the inquiry, staff manually checks vehicle availability using an internal tracking system. This depends entirely on staff accuracy and real-time updates. If a car is available, customers are required to make either a deposit or full payment before the booking can be confirmed.

2.1.2 Customer Registration and Verification

Customers provide personal details such as identification cards or passports and driving licenses via WhatsApp. Staff manually verifies these documents and stores them in their internal systems. Once full payment is completed, staff generates the rental agreement and sends it to the customer. Customers then digitally agree to the terms and conditions through a signature or checkbox with a timestamp. However, this still requires manual handling and verification by staff that risks the records consistency.

2.1.3 Payment Process

Payments are accepted through QR code, online bank transfer, or cash and proof of payment is sent in the form of a receipt screenshot. Staff manually verifies each transaction and updates the payment status in the system. A centralised system for this is necessary to log all the payment logs and document the verification to prevent error.

2.1.4 Car Collection and Return

On the rental day, customers collect the vehicle at the office location. Staff prepares a checklist to document the car's condition, fuel level, and other relevant details during handover. Upon return, staff inspects the vehicle for scratches, dents, cleanliness, fuel level, and overall condition. The return inspection details are then recorded in the system. There is no proper tracking system to calculate late return penalties or to notify customers in real time which increases the risk of delays and customer disputes.

2.1.5 Reporting

Rental records, payment details, vehicle usage, and return data are compiled using a combination of the Track system and Microsoft Excel. Management reports, such as monthly sales summaries, rental history, and car utilization reports, are prepared manually at the end of each month. Compiling data from multiple sources manually is time-consuming and it could lead to data inconsistencies.

3.0 Problem Statement

Currently, the booking management system that is used is manual, contacting customers through Whatsapp which is inconvenient for the staff to keep record of bookings and customer information. Hasta is also using Driveo for car rental management but it is not customized to the needs of Hasta and their target audience which is UTM students and staff.

In addition, lack of a centralized data storage is a weakness to the current system since customer information, booking details, vehicle records and payment records are scattered that leads to data inconsistency. This is fatal for the operation of Hasta for their audit and other legal purposes. Without a centralised system, the workload of staff is huge and time consuming.

The current system doesn't have automated analytics and reporting generated on bookings, vehicle usage, customer demographics, outreaches, and business performance. This is essential for businesses to make crucial decisions to expand their reach and come up with effective solutions to boost staff performances.

Furthermore, the current system lacks a structured method of handling customer issues. The complaints and service requests aren't properly tracked and recorded, making it difficult for staff to promptly fulfill the requests. This has an effect on the customer experience that determines the customer loyalty to the company.

Lastly, the current system is not scalable enough and cannot support if the number of customer, vehicles, bookings increases. Since it relies heavily on manual processes and staff intervention, staff has to spend more time checking availability, updating records and preparing reports which increases the workload and decreases the efficiency of staff.

4.0 Proposed Solution

4.1 Feasibility Studies

A feasibility study was conducted to evaluate the viability of developing the GoRent online car rental system for Hasta Travel Tours Sdn. Bhd. The purpose of this system is to digitalize the company's current car rental system, which is still largely manual, and create a more convenient and efficient booking experience for UTM students and staff. The study analyzed three key areas: technical, operational, and economic feasibility. This section covers the findings related to technical and operational aspects.

4.1.1 Technical Feasibility

From a technical overview, the proposed GoRent system is highly feasible. Hasta Travel already has an existing website called hastatravel.com that can serve as a foundation for system integration and expansion. Although the current site lacks interactive features, modern web technologies such as HTML5, JavaScript, PHP (or Laravel), and MySQL make it fully possible to develop a dynamic system with real-time booking, car listing, and online payment capabilities. Furthermore, the company operates within Universiti Teknologi Malaysia (UTM), which provides a reliable internet infrastructure for an ideal hosting and maintaining a web-based system. With access to skilled developers, possibly in collaboration with university students or IT professionals, the technical implementation of the system is both achievable and sustainable.

4.1.2 Operational Feasibility

In terms of operations, the GoRent system aligns well with Hasta Travel's existing business model and processes. The system will streamline the manual workflow of handling car rental inquiries by automating tasks such as booking management, vehicle availability tracking, customer registration, and receipt generation. This automation not only reduces the workload of administrative burdens on staff but also enhances the overall customer experience through faster response times and improved service accuracy. The target audience are mainly UTM students and staff that are already familiar with using online platforms, which suggests a high likelihood of successful system usage. Additionally, the system can be scaled in the future to support more services, such as tour package bookings and loyalty programs, further enhancing operational efficiency and service delivery.

4.1.3 Economical Feasibility

The GoRent system is economically feasible as it requires relatively low development and maintenance costs due to the use of existing infrastructure and open-source technologies. The system reduces operational costs by minimizing manual work and improving staff efficiency. Increased booking convenience and customer satisfaction may also lead to higher revenue, making the system financially beneficial in the long run.

5.0 Objectives

The aim of this system is to build a centralized, secure, and easy-to-use web-based management system specifically for UTM students and staff. This system is designed to digitalize and improve the entire car rental process for Hasta Travels & Tours. It will help automate manual tasks such as booking, deposit payment, agreement signing, car inspection, loyalty tracking, and payment handling, so the operations become more accurate and transparent for both customers and staff. The system also aims to support the company's growth by providing useful analytics, real-time activity updates, and clear visibility of car conditions, booking history, and customer behaviour.

Objectives :

- The first mission objective is to improve the current car rental process at Hasta Travels & Tours by changing the manual workflow into a centralized web-based system. This system helps reduce errors such as double bookings and missing records by automating booking, deposit tracking, and agreement signing. As a result, staff can manage rentals more efficiently and with less manual work.
- The second objective is to increase transparency and proper record management for both customers and staff. All booking details, payments, inspection photos, and customer histories are stored in one system. This allows customers to easily check their booking status, deposits, refunds, and penalties, while staff can track car usage and customer records more clearly.
- Another objective is to provide a better and more convenient experience for UTM students and staff. Customers can check car availability in real time, make bookings online, and view their rental history without needing to visit the office. Features such as weekday promotional pricing and loyalty rewards are included to make the service more attractive and user-friendly.
- The system also aims to help management make better decisions by providing simple reports and analytics. Information such as booking frequency, revenue, and customer trends by residential colleges allows Hasta Travels & Tours to understand their business performance and plan improvements more effectively.
- Lastly, this system is designed to support future business growth. It can easily handle more users, vehicles, and new features when needed. This ensures the system remains useful and suitable for long-term use as Hasta Travels & Tours expands its services.

6.0 Scope and Boundaries

Project Scope

This project covers the design and development of a Web-Based Car Rental Management System for Hasta Travel & Tours. The system will support both customers and administrative staff by automating core rental operations and improving service efficiency. The key functionalities of the project scope are:

1. User Registration and Verification

- Registration for UTM students and staff using valid institutional credentials.
- Upload of identification documents and driving licenses for manual verification by administrators.
- Customer profile management including loyalty points and deposit balance tracking.
- Mandatory agreement to system terms and policies during registration.

2. Login and Blacklist Control

- Access is granted only to verified users.
- Blacklisted users are restricted from logging in until outstanding penalties or overdue payments are settled.
- Tracking and management of multiple blacklist reasons such as unpaid fines and repeated policy violations.

3. Car Management

- Add, edit, deactivate, and manage vehicles in the fleet.
- Record pickup and drop-off locations..
- Automatic update of car status when mileage exceeds the service limit.

4. Booking Management

- Customers can browse and select available cars.
- Full payment booking process (Phase 1) with a default deposit of RM50 per booking.
- Option to retain or refund deposit after booking (Phase 3).
- Mandatory upload of before and after rental photos.
- Customers must upload key handover photos during pickup.
- Digital signing of rental agreements.

- Cancellation and rescheduling allowed only one day before the booking date.

5. Payment and Penalty Handling

- Manual payment via QR code and bank transfer with receipt upload.
- Recording of deposits, rental payments, penalties, and refunds.
- Automatic penalty calculations including:
 - Late return fee: RM30 per hour
 - Fuel shortage penalty: RM10 per bar
 - Damage and cleanliness penalties based on inspection
- Deposit refund or retention logic.
- Monthly and ad-hoc penalty reporting.
- Support for installment plans for accident/damage penalties.
- Optional monthly installment payments for long-term rentals (6 months and above).
- Deposit refunds processed within 10 working days.
- Receipt uploads require only payment amount and image proof.
- Loyalty-based discount calculation.

6. Loyalty Rewards

- Tracking rental hours (1 stamp per 9 hours).
- Admin create voucher generation.
- Voucher redemption during checkout.
- Display of available vouchers on customer dashboard.

7. Admin and Staff Functions

- Dashboard with real-time booking, revenue, and activity statistics.
- Pre- and post-rental car inspections.
- Update vehicle status (available, in use, maintenance).
- Manage customer verification and blacklist records.
- View booking frequency by residential college.
- Generate financial and operational reports.

8. Feedback and Issue Reporting

- Customers can submit feedback and report vehicle issues after each completed booking.

- Admin and staff can review feedback and link issues to maintenance actions.

Project Boundaries

The following features are excluded from the project scope:

1. Vehicle Insurance Processing

- No handling of insurance claims or documentation.

2. Integration with External Banking APIs

- No FPX, ToyibPay, Stripe, or other payment gateway integrations.
- All payments rely on manual receipt uploads.
- Admin need to verify the receipt(requested by the stakeholder)

3. Vehicle Maintenance Scheduling

- The system only notifies when mileage exceeds service limits.
- No automatic maintenance scheduling functionality.

4. Automated Staff Assignment

- Staff task allocation and scheduling remain manual.

6.1 Use Case Diagram

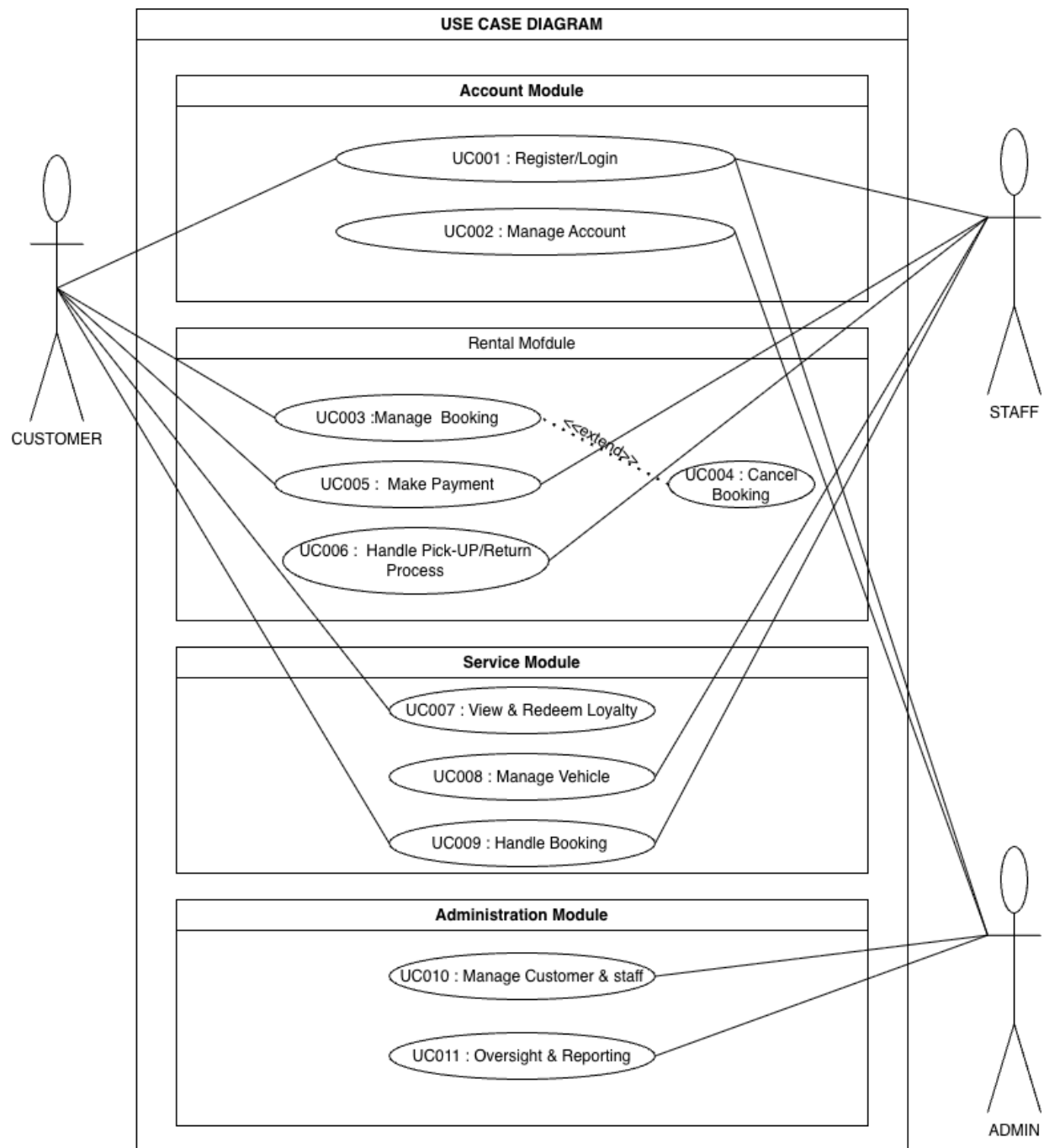


Figure 1.1: Use Case Diagram for Hasta Travel & Tours Car Rental System.

7.0 Project Planning

7.1 Human Resource

A committed project team with distinct roles and responsibilities is necessary for the development and successful implementation of Hasta Travel & Tours Sdn. Bhd.'s car rental system. To make sure the system satisfies the needs of the client, each team member brings specialised skills to the planning, designing, development, and testing phases.

1.The project manager is in charge of managing the schedule, assigning tasks to team members, keeping an eye on the entire undertaking, and making sure that all deadlines are fulfilled. Additionally, the project manager acts as the team's primary point of contact with the client (Hasta Travel).

2.The system analyst compiles user needs, evaluates the manual system that is in place, and establishes the requirements for the system. The analyst makes sure the suggested solution fits Hasta Travel's operational requirements.

3.UI/UX designer creates the system's layout, wireframes, and user interface to guarantee that administrators and customers have an easy-to-use and intuitive experience.

4.Programmers are in charge of coding the system in accordance with the specifications and design. Implementing front-end and back-end features like database integration, admin dashboards, and booking modules is part of this.

5.Database administrators create and oversee the database architecture of the system, guaranteeing effective data security, retrieval, and storage. If necessary, the DBA also manages data migration from current manual records.

6.A quality assurance tester checks the system for errors, malfunctions, and issues with the user experience. The system's stability, security, and deployment readiness are guaranteed by QA.

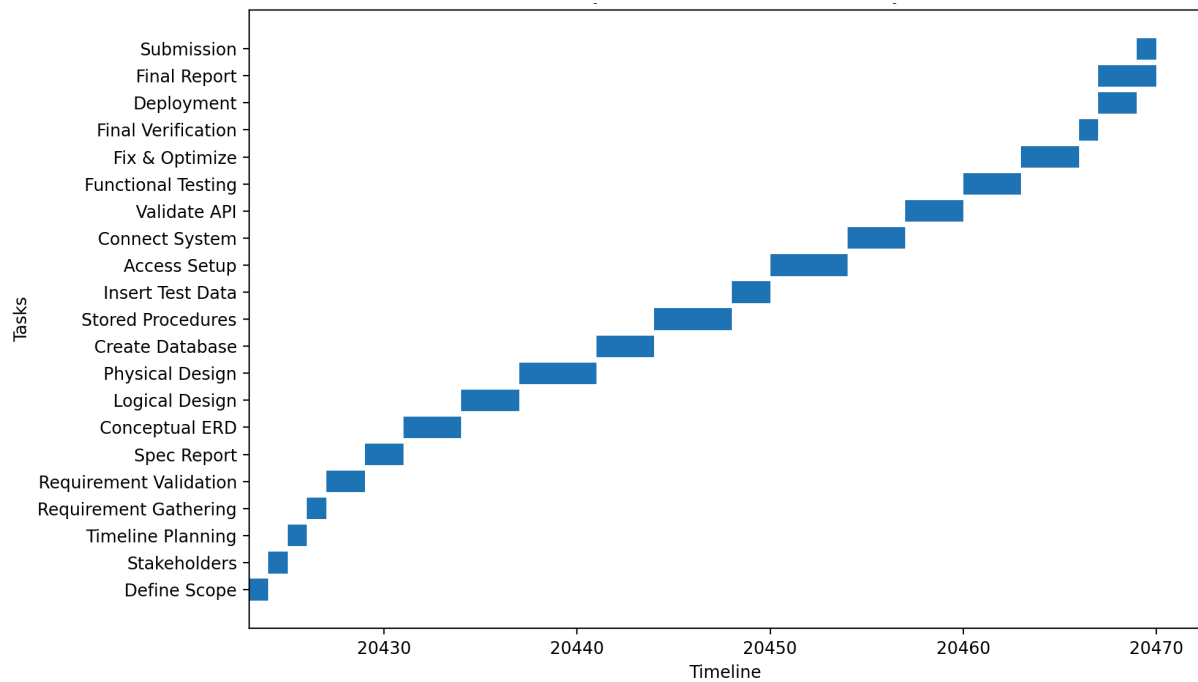
7.Technical Support/Deployment Officer: Helps install the system, deploy it to a live server, and train or document the new system for Hasta Travel employees.

7.2 Work Breakdown Structure (WBS)

Product-Oriented WBS – Car Rental System

No	Task Description	Assigned To	Start	End
1.0	Project Planning			
1.1	Define Project Scope & Objectives	Kavinesh,	1/12/2025	1/12/2025
1.2	Identify Stakeholders & Constraints	Nive	2/12/2025	2/12/2025
1.3	Prepare Project Timeline & Task Allocation	al,haykal	3/12/2025	3/12/2025
2.0	Requirement Analysis			
2.1	Requirement Gathering from Stakeholders	kavinesh, al	4/12/2025	4/12/2025
2.2	Requirement Validation & Finalization	nive, kal	5/12/2025	6/12/2025
2.3	Prepare Requirement Specification Report	kavinesh	7/12/2025	8/12/2025
3.0	System & Database Design			
3.1	Conceptual ERD Design	Kavinesh	9/12/2025	11/12/2025
3.2	Logical Design - Table Structure	al	12/12/2025	14/12/2025
3.3	Physical Design - Normalization, Constraints, Indexing	haykal	15/12/2025	18/12/2025
4.0	Database Deployment			
4.1	Create Database & Tables	nive	19/12/2025	21/12/2025
4.2	Develop Stored Procedures & Triggers	al	22/12/2025	25/12/2025
4.3	Insert Test Data	kavinesh	26/12/2025	27/12/2025
4.4	Admin/Staff Access Setup	kal	28/12/2026	31/12/2026
5.0	Integration & System Configuration			
5.1	Connect System to Database	kavinesh,al	1/1/2026	3/1/2026
5.2	Validate API/Query Interactions	nive,kal	4/1/2026	6/1/2026
6.0	Testing & Quality Assurance			
6.1	Functional Testing – Queries, Transactions	kavinesh, nive	7/1/2026	9/1/2026
6.2	Error Fixing & Optimization	All	10/1/2026	12/1/2026
6.3	Final System Verification	al	13/1/2026	13/1/2026
7.0	Deployment & Documentation			
7.1	Final Deployment & Backup Setup	Kavinesh,	14/1/2026	15/1/2026
7.2	Final Report, Diagrams & User Guide	nive	14/1/2026	16/1/2026
7.3	Project Submission	al,kal	16/1/2026	16/1/2026

7.3 Gantt Chart



8.0 Requirement Analysis (Based From As-Is Analysis)

The requirement analysis is conducted to understand the current manual operations of Hasta Travel & Tours. By analyzing the "As-Is" system, we can identify the functional needs for the proposed GoRent system to ensure it solves the existing operational problems effectively.

8.1 Current Business Process

Currently, Hasta Travel & Tours relies heavily on manual methods to manage their car rental business. The business process flows as follows:

- **Car Inquiry and Reservation:** The process starts when a customer makes an inquiry through WhatsApp, phone calls, or walk-ins. Staff must manually check car availability by referring to spreadsheets or written records.
- **Customer Registration and Verification:** Customers are required to submit copies of their identification cards (IC), passports, and driving licenses via WhatsApp. Staff will then manually verify these documents and store them in their internal systems.
- **Payment Process:** Payments are made via QR code, cash, or online bank transfer. Customers must provide a screenshot of the receipt as proof of payment, which staff then verifies and records manually in the system.
- **Car Collection and Return:** During pickup, staff uses a checklist to record the car's condition and fuel level. Upon return, staff performs a manual inspection for any damages or fuel shortages. Currently, there is no automated system to track late return penalties or notify

customers in real-time.

- **Reporting and Management:** At the end of each month, rental records, payment details, and vehicle usage data are compiled manually using Microsoft Excel and the existing "Track" system. This manual compilation is time-consuming and often leads to data inconsistencies.

9.0 Transaction Requirement (Data Entry, Data Update/Delete, Data Queries)

9.1 Data Entry

- **Booking Info:** Customers will fill in their booking details, including rental dates, car choice, personal info (like ID and contact), and payment method on the website or app.
- **Customer Info:** Customers will provide personal details like ID, contact info, and any past rental history when they register or book a car.
- **Payment Info:** Payment details such as the amount, method (QR code, bank transfer), and proof of payment (receipt image) will be entered for each booking.
- **Car Condition:** Staff will add details like the car's fuel level, condition (e.g., scratches, dents), mileage, and any issues at pickup or return.

9.2 Data Update/Delete

- **Customer Profile:** Customers or staff can update contact details or upload new documents for verification. Admins can remove outdated or incorrect profiles.
- **Booking Changes:** If there are changes (like date or car changes), the booking info will be updated. If someone cancels, the booking record will be deleted or marked as canceled.
- **Payment Status:** Any updates to payments (like successful payments, refunds, or penalties) will be updated in the system. Wrong or duplicate payments can be deleted if found.
- **Car Info:** Car details, like availability or condition, can be updated. If a car is no longer in the fleet, it will be removed from the system.

9.3 Data Queries

- **Booking Search:** Admins can search for booking details like available cars, booking status, or customer history. They can search by customer name, date, car type, or payment status.

- **Payment Search:** Admins can look up payment records to check for any pending payments, receipts, refunds, or completed transactions.
- **Customer Search:** Admins can search for customer details like past bookings, rental preferences, and loyalty points.
- **Vehicle Info Search:** Admins can check the status of cars, including current bookings, mileage, and last service date.
- **Reports:** Admins can generate monthly or ad-hoc reports on sales, penalties, and bookings. They can filter reports based on dates, car types, and customer groups.

10.0 Benefit and Summary of Proposed System

The proposed GoRent Car Rental Management System offers a comprehensive digital solution to address the current inefficiencies in vehicle rental services within the university environment. By introducing a centralized, web-based platform, the system streamlines booking, payment, vehicle inventory, and customer management processes. This automation reduces administrative burden, enhances service efficiency, and increases transparency through electronic records and real-time updates.

Users, both students and staff, benefit from convenient 24/7 access to car availability, secure payments, and verified registration, ensuring smoother transactions and higher user satisfaction. On the administrative side, the system provides detailed reporting and analytics, enabling data-driven decision-making and future scalability. Financial projections indicate long-term profitability, with a profitability index of 2.12 by Year 3, proving that the system is both viable and beneficial.

Overall, GoRent stands as a sustainable, scalable, and modern transport solution tailored for the university's needs.

11.0 Summary

This report proposes a web based car rental system for Hasta Travel & Tours to replace the current manual system. The proposed system is suggesting to solve the issue of data redundancies, manual booking errors and tedious processes. With a centralised database there is more credibility and security for the data stored which helps to reduce human error. The proposed system enhances customer experience by enabling real-time car availability, providing the facility to book online that is seamless and efficient. The staff benefit from the proposed system with easier payment tracking, structured reporting and analytics, effective management of bookings, penalties and blacklists. The proposed system is also feasible in terms of technical, operational and economical aspects which is practical and efficient for Hasta Travels & Tours. Thus, the proposed system is reliable and suitable to be adapted by Hasta Travels & Tours to improve their service quality and overall performance.

